

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Code of Conduct:

I S. Nitheesh (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S. Nitheesh

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Scenario	20%				
Identification of Key Issues	20%				
Application of Concepts	25%				
Decision Making	20%				
Clarity of Response	15%				

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.

2. The matching routing table entry.
3. The next-hop network/interface used to forward the packet.
4. The default route that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → 192.168.10.0/26

Finance LAN → 192.168.10.64/26

The router interfaces are configured with:

Administration Gateway → 192.168.10.1

Finance Gateway → 192.168.10.65

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario

		g of the scenario and context			
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned , logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganize d and difficult to understand

Subnetting of 192.168.20.0/24 Network

Given :

IP Address Block - 192.168.20.0/24

Lab A : /26

Lab B : /27

Lab C : /28

Step 1 : Lab A (/26)

Subnet Mask: 255.255.255.192

Number of Host Bits : $32 - 26 = 6$

Total IP Addresses : $2^6 = 64$

Usable Host Addresses : $64 - 2 = 62$

Network Address : 192.168.20.0

First Host Address : 192.168.20.1

Last Host Address : 192.168.20.62

Broadcast Address: 192.168.20.63

Step 2: LAB B (/27)

Subnet Mask: 255.255.255.224

Number of Host Bits : $32 - 27 = 5$

Total IP Addresses : $2^5 = 32$

Usable Host Addresses : $32 - 2 = 30$

Next available network: 192.168.20.64

Network address: 192.168.20.64

First Host: 192.168.20.65

Last Host: 192.168.20.94

Broadcast Address: 192.168.20.95

Step 3: Lab C (128)

Subnet Mask: 255.255.255.240

Number of Host Bits: $32 - 28 = 4$

Total IP Addresses: $2^4 = 16$

Usable Host Addresses: $16 - 2 = 14$

Next available network: 192.168.20.96

Network Address: 192.168.20.96

First Host: 192.168.20.97

Last Host: 192.168.20.110

Broadcast Address: 192.168.20.111

Subnet Analysis of Two Systems :

Given :

System 1 IP Address : 192.168.1.130

System 2 IP Address : 192.168.1.190

Subnet Mask : 255.255.255.192 (126)

Step 1: Calculate the subnet information.

Subnet Mask = 255.255.255.192

Host Bits = $32 - 26 = 6$

Total IP Addresses = $2^6 = 64$

Usable Host Addresses = $64 - 2 = 62$

Subnet increment = 64

Subnets are:

192.168.1.0 - 192.168.1.63

192.168.1.64 - 192.168.1.127

192.168.1.128 - 192.168.1.191

192.168.1.192 - 192.168.1.255

System 1 Analysis

IP Address = 192.168.1.130

Network Address = 192.168.1.128

First Host = 192.168.1.129

Last Host = 192.168.1.190

Broadcast Address = 192.168.1.191

System 2 Analysis:

IP Address = 192.168.1.190

Network = 192.168.1.128

First Host = 192.168.1.129

Last Host = 192.168.1.190

Broadcast Address = 192.168.1.191

Subnetting of 172.16.0.0/16

Given:

Network : 172.16.0.0/16

HR : 125

Sales : 126

IT : 127

Admin : 128

① HR Department (125)

Network Address : 172.16.0.0

First Host : 172.16.0.1

Last Host : 172.16.0.126

Broadcast Address : 172.16.0.127

Usable Hosts : 126

② Sales Department (126)

Network Address : 172.16.0.128

First Host : 172.16.0.129

Last Host : 172.16.0.190

Broadcast Address : 172.16.0.191

Usable Hosts : 62

③ Admin Department (128)

Network Address: 172.16.0.224

First Host: 172.16.0.225

Last Host: 172.16.0.238

Broadcast Address: 172.16.0.239

Usable Hosts: ~~20~~ 14

④ IT Department (127)

Network Address: 172.16.0.192

First Host: 172.16.0.193

Last Host: 172.16.0.222

Broadcast address: 172.16.0.223

Usable Hosts: 30

Subnetting Using /27:

Given:

Network: 172.16.10.0/24

Subnet Mask: /27

Step 1: Calculate Number of Subnets:

Origin Prefix = /24

New Prefix = /27

Borrowed Bits = $27 - 24 = 3$

Number of subnets = $2^3 = 8$

Step 2: Calculate Usable Hosts:

Host Bits = $32 - 27 = 5$

Total Addresses = $2^5 = 32$

Usable Hosts = 30

Step 3: Identify the subnet of 172.16.10.78

172.16.10.0 - 31

172.16.10.32 - 63

172.16.10.64 - 95

172.16.10.96 - 127

172.16.10.128 - 159

172.16.10.160 - 191

172.16.10.192 - 223

172.16.10.224 - 255

Therefore;

Network Address = 172.16.10.64

First Host = 172.16.10.65

Last Host = 172.16.10.94

Broadcast Address = 172.16.10.95

Step 4: Verify 172.16.10.95

The IP address 172.16.10.95 is the broadcast address of the subnet. Therefore, it cannot be assigned to a host

IPv6 Addressing:

Given IPv6 Address:

2001:0db8:0000:0000:0000:ff00:0042:8329

Step 1: Compress the IPv6 Address:

Rules:

- * Remove leading zeros.
- * Replace one continuous sequence of zero blocks with "::".

Compressed Address:

2001:db8::ff00:42:8329

Step 2: Expand the IPv6 Address

Expanded Address:

2001:0db8:0000:0000:0000:ff00:0042:8329

Step 3: Determine the Network Prefix

Prefix Length = 64

Network Prefix = 2001:0db8:0000:0000::/64

The first 64 bits represent the network portion, and the remaining 64 bits represent the interface identifier.

Step 1: Calculate the number of Host Addresses

$$\text{Host Bits} = 128 - 64 = 64$$

Number of Host Addresses:

$$2^{64} = 18,446,744,073,709,551,616$$

This extremely large address space ensures that every device can have a unique IP address